SAR: State Aware Retrieval for Iterative Retrieval Augmented Generation

A Synopsis report submitted in

partial fulfillment of requirements for the degree of

Bachelor of Technology

In

Computer Engineering

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2026

ABSTRACT

Retrieval-Augmented Generation (RAG) combines information retrieval with large language models so that answers can be supported by information from external documents. This is especially useful for complex questions whose required evidence may be distributed across multiple documents. However, when retrieval is performed iteratively, selecting documents only on the basis of their relevance to the original question may cause the system to retrieve information that has already been covered instead of information that is still required.

This project is going to study and develop a **Gap-Aware Retrieval (GAR)** approach for iterative RAG. The proposed approach is going to consider the current question, the information that is still missing, and the evidence that has already been collected when selecting documents for the next retrieval step. The project is going to focus on reducing redundant retrieval and improving the selection of complementary evidence for complex question answering. A mathematical formulation for the retrieval state and a document-scoring equation is going to be developed as part of the project; no final equation has been established at the synopsis stage. The proposed retrieval approach is then going to be integrated into the existing iterative RAG setting and evaluated through systematic experiments.

Keywords: Retrieval-Augmented Generation, Information Retrieval, Gap-Aware Retrieval, Iterative Retrieval, Missing Information, Evidence Retrieval, Multi-Hop Question Answering.

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1. INTRODUCTION

Retrieval-Augmented Generation (RAG) is a framework in which a language model retrieves relevant information from an external document collection and uses that information to generate an answer. By supplying external evidence at generation time, RAG extends a language model beyond the information contained only in its parameters and is therefore useful for knowledge-intensive question answering [1]. The central role of the retrieval stage is to select a small set of documents that can provide the evidence required to answer the question [1].

The first additional work considered in this project is the work of Izacard and Grave, which uses multiple retrieved passages with a generative model. Its findings show that an answer can depend on useful information distributed across several retrieved passages rather than on a single passage alone [2]. This indicates that retrieval quality is not only about finding one highly relevant document, but also about obtaining a useful collection of evidence [2]. Yadav et al. further studied iterative evidence retrieval for multi-hop question answering. Their work shows that later retrieval steps can be guided by information that has not yet been covered by the evidence already collected [3]. This suggests that the retrieval requirement can change after each retrieval step [3].

Wang et al. introduced MIGRES, a missing-information-guided iterative framework in which the system identifies what information is still missing after evidence has been collected and uses that information to guide the next retrieval step [4]. Thus, the retrieval process can maintain an evolving state consisting of what is already known and what is still required [4]. Pickett et al. studied Relevant Information Gain and emphasized selecting information that adds useful content while avoiding unnecessary redundancy with information already retrieved [5]. Their work therefore provides a basis for considering the amount of new information contributed by a candidate document [5].

From these findings, the problem discovered in the present project is that a conventional relevance-based retrieval decision may remain focused on the original question even after part of the required information has already been obtained. In an iterative setting, this can cause later retrieval steps to return documents that are related to the question but overlap with existing evidence, while documents containing complementary or still-missing information may receive less attention. The issue is therefore not simply whether a document is relevant to the question, but whether it is useful with respect to the current information gap and the evidence already collected.

To address this problem, the project is going to introduce **Gap-Aware Retrieval (GAR)** as a reranking component within the iterative RAG setting. The project is going to use the existing MIGRES process to provide the current information gap and previously collected evidence, while GAR is going to focus on selecting the most useful documents from the retrieved candidate set. The proposed approach is going to consider three directions: relevance to the question and current gap, coverage of the missing information, and novelty with respect to previous evidence. A mathematical formulation for combining these signals is going to be developed during the project; no final equation is being claimed at the synopsis stage.

The proposed approach is expected to reduce redundant retrieval and help later retrieval steps focus on complementary evidence that is still needed. It is also going to be evaluated for its effect on retrieval quality and downstream question answering within the iterative RAG pipeline.

CHAPTER 2

LITERATURE SURVEY

Table 2.1: Literature Survey 1

Name of paper	Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks
Authors	Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela
Publication Details	Advances in Neural Information Processing Systems 33 (NeurIPS), 2020, pp. 9459–9474.
Research Findings	• RAG combines information stored in a language model with information retrieved from external documents. • Retrieved passages help the model answer questions that require outside knowledge. • The approach shows that retrieval can improve the performance of language generation for knowledge-intensive tasks. • The work provides a basic foundation for modern retrieval-augmented generation systems.
Research Gap	• The main retrieval process is based on finding passages relevant to the given question. • It does not mainly focus on identifying which information is still missing after previous evidence has been retrieved. • The work does not specifically study how retrieval can change according to the information already collected.
Relevance with Our Work	• This paper provides the basic RAG background for our project. • It supports the use of external documents to provide additional information to language models. • Our work builds on this general retrieval setting by studying retrieval in a more iterative information-seeking process.

Table 2.2: Literature Survey 2

Name of paper	Leveraging Passage Retrieval with Generative Models for Open Domain Question Answering
Authors	Gautier Izacard and Edouard Grave
Publication Details	Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics (EACL), 2021, pp. 874–880.
Research Findings	• The method retrieves multiple passages and provides them to a generative model. • The model can use information from several passages to answer a question. • Using useful information from multiple retrieved passages can improve question-answering performance. • The work shows the importance of selecting and using several relevant pieces of evidence.
Research Gap	• The focus is mainly on how a generative model uses multiple retrieved passages. • It does not mainly study how the retrieval process should identify information that is missing from earlier evidence. • The retrieved passages are not selected according to a changing information state across multiple retrieval steps.
Relevance with Our Work	• This work supports the importance of collecting multiple useful pieces of evidence. • It is indirectly related to our goal of improving the usefulness of retrieved documents. • Our project further considers how the current information need can be used when selecting documents during iterative retrieval.

Table 2.3: Literature Survey 3

Name of paper	LLMs Know What They Need: Leveraging a Missing Information Guided Framework to Empower Retrieval-Augmented Generation
Authors	Keheng Wang, Feiyu Duan, Peiguang Li, Sirui Wang, and Xunliang Cai
Publication Details	Proceedings of the 31st International Conference on Computational Linguistics (COLING), 2025, pp. 2379–2400.
Research Findings	• MIGRES identifies information that is still missing after processing the retrieved evidence. • The identified missing information is used to guide the next retrieval step. • The framework performs retrieval and evidence processing in multiple steps. • The approach is designed for questions that require several pieces of information.
Research Gap	• The framework mainly focuses on identifying missing information and guiding the next retrieval query. • There is still room to study how candidate documents can be ranked using both the missing information and the evidence already collected. • Different document-ranking strategies may affect which evidence is selected during the iterative process.
Relevance with Our Work	• MIGRES provides an important background for understanding missing information during iterative RAG. • It supports the idea that later retrieval should focus on information that is still required. • Our project uses this general idea as motivation for studying a gap-aware retrieval and reranking approach. • The paper provides the main framework in which our proposed approach can be investigated.

Table 2.4: Literature Survey 4

Name of paper	Unsupervised Alignment-based Iterative Evidence Retrieval for Multi-hop Question Answering
Authors	Vikas Yadav, Steven Bethard, and Mihai Surdeanu
Publication Details	Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL), 2020, pp. 4514–4525.
Research Findings	• AIR is an iterative method designed to retrieve evidence for multi-hop questions. • The retrieval process changes after each step instead of using the same query throughout. • Later retrieval focuses on information that is not already covered by the available evidence. • The work shows that iterative retrieval can help collect complementary evidence.
Research Gap	• The method mainly focuses on changing the retrieval query based on previously collected information. • It does not fully explore a separate ranking stage that considers relevance, missing information, and existing evidence together. • There is room to investigate other ways of selecting complementary documents during iterative retrieval.
Relevance with Our Work	• AIR is closely related to the idea of avoiding already-covered information. • It supports the importance of retrieving complementary evidence in multi-step question answering. • This provides useful background for studying a retrieval method that considers the current information state.

Table 2.5: Literature Survey 5

Name of paper	Better RAG using Relevant Information Gain
Authors	Marc Pickett, Jeremy Hartman, Ayan Kumar Bhowmick, Raquib-ul Alam, and Aditya Vempaty
Publication Details	arXiv preprint, 2024, arXiv:2407.12101.
Research Findings	• The paper studies how relevant information gain can be used to improve document selection in RAG. • It aims to reduce redundant information while keeping the retrieved information relevant. • The approach gives importance to documents that provide useful information not already present in the selected evidence. • The work highlights the importance of balancing relevance with the usefulness of new information.
Research Gap	• The main focus is on selecting useful and non-redundant information from retrieved passages. • It does not mainly focus on a changing information need across several retrieval steps. • The current missing information can provide an additional signal for deciding which document should be selected next.
Relevance with Our Work	• This work provides useful background for reducing repeated or unnecessary evidence. • It supports the idea that document selection should consider more than simple relevance. • The concept of useful new information is indirectly related to our goal of selecting documents that help cover the current information gap.

CHAPTER 3

PROBLEM DEFINITION

1. In iterative Retrieval-Augmented Generation (RAG), some useful information may already be retrieved in earlier steps. However, the next retrieval may return similar information instead of focusing on what is still missing.
2. In conventional retrieval, documents are mainly selected according to their relevance to the query. This may favor information that has already been collected instead of information that is still missing.
3. Previous evidence is not fully considered when selecting documents for later retrieval. As a result, redundant documents may be selected while useful and complementary documents may be missed.
4. Therefore, there is a need for a state aware retrieval approach that considers the query, missing information, and previously collected evidence to select more useful documents for the next retrieval step.

CHAPTER 4

OBJECTIVES

1. To study existing RAG and iterative retrieval methods and identify the limitations of relevance-based retrieval in later retrieval steps.
2. To study how missing information can be used to guide document retrieval in an iterative setting.
3. To study how previously collected evidence can be used to reduce redundant information and select complementary documents.
4. To experimentally evaluate and verify the effectiveness of the proposed Gap-Aware Retrieval (GAR) for improving document retrieval in iterative RAG.

CHAPTER 5

APPLICATIONS

1. **Complex Question Answering:** Can support AI systems in answering complex questions where the required information is spread across multiple documents and cannot be obtained through a single retrieval step.

2. **Multi-Step Information Retrieval:** Can be applied to systems that repeatedly search for information, helping each retrieval step focus on information that has not yet been obtained.

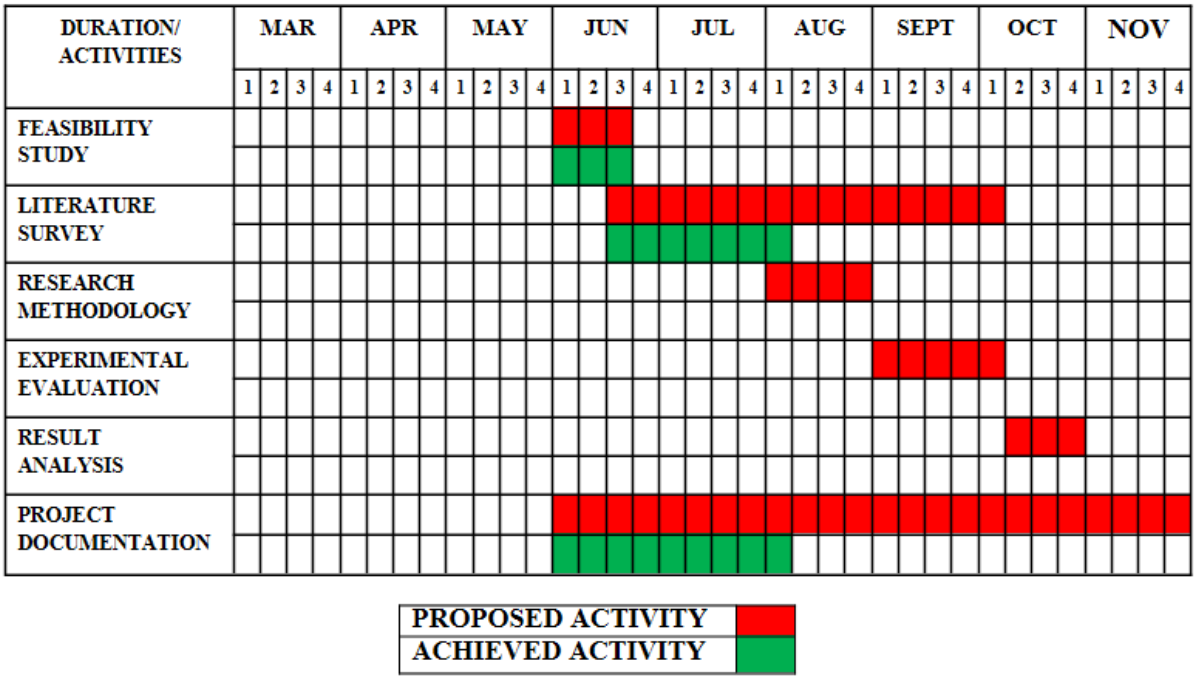
3. **Research and Knowledge Discovery:** Can assist in collecting information from large document collections by prioritizing useful and complementary evidence instead of repeatedly retrieving similar content.

4. **Evidence-Based RAG Applications:** Can be integrated into RAG-based applications where the system needs to build a complete set of evidence before generating an answer, such as question answering, document analysis, and knowledge-based assistants.

CHAPTER 6

GANTT CHART

Figure 6.1: Gantt Chart



CHAPTER 7

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